THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · EASY LEVEL CHAPTER TEST SET 1

Chapter 10: **How Quantities Combine (Understanding Data)** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The mean of 44, 33, 40, 35, 58 is: 1

(a) 42 **(b)** 44 **(c)** 43 **(d)** 41

2. Over 5 weeks of equal length a cyclist's weekly average distances were 32, 54, 42, 49, 23 km per day. The average over the whole period (km per day) is: 1

(a) 42 **(b)** 39 **(c)** 40 **(d)** 41

3. In a 100% stacked bar of a household's monthly electricity use, appliances take 29% of 300 units. The units used by the appliances are: 1

(a) 86 **(b)** 87 **(c)** 89 **(d)** 88

4. A stacked bar for a family's monthly food use shows rice 32 kg, wheat 24 kg, pulses 39 kg. The total length of the bar is (in kg): 1

(a) 95 **(b)** 94 **(c)** 96 **(d)** 97

5. The average marks of 24 students is 46 and of another 15 students is 59. The average of all 39 students is: 1

(a) 53 **(b)** 51 **(c)** 50 **(d)** 52

6. The concentration of juice in a drink is 12%. The amount of juice in 250 mL of the drink is: 1

(a) 3 mL **(b)** 25 mL **(c)** 30 mL **(d)** 12 mL

7. When two groups of different sizes are combined, the combined average is found by: 1

(a) adding the two averages and halving
(b) subtracting the smaller average from the larger
(c) total of all values divided by the total number of values
(d) multiplying the two averages

8. Marks of 79, 64 and 69 are given weights 2, 1 and 2 respectively. The weighted average is: 1

(a) 73 **(b)** 72 **(c)** 71 **(d)** 74

9. The mean of 22, 48, 47, 26, 36, 19 is: 1

(a) 35 **(b)** 32 **(c)** 34 **(d)** 33

10. Over 5 weeks of equal length a cyclist's weekly average distances were 52, 33, 58, 34, 38 km per day. The average over the whole period (km per day) is: 1

(a) 42 **(b)** 44 **(c)** 43 **(d)** 45

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the mean of the numbers 55, 27, 50, 44, 54. 2

12. A 100% stacked bar shows that 25% of a family's monthly spending of ₹200 is on food. Find the amount spent on food and on everything else. 2

13. A 250 mL bottle of a drink contains 36% fruit pulp. Find the volume of pulp and of the remaining liquid. 2

14. Find the mean of the numbers 44, 36, 42, 30. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The average marks of 10 students in a test is 47 and that of 30 students is 31. Find the average marks of all 40 students together. 3

16. A girl cycles for 3 weeks of equal length. Her weekly averages are 56 min, 26 min, 41 min per day. Find the average over all the weeks and say why the simple mean of the averages can be used. 3

17. A final score combines Maths (score 83, weight 3), Science (score 61, weight 3). Find the weighted average score. 3

OR

The average marks of 24 students in a test is 46 and that of 15 students is 59. Find the average marks of all 39 students together. 3

18. A girl cycles for 5 weeks of equal length. Her weekly averages are 53 min, 53 min, 38 min, 57 min, 24 min per day. Find the average over all the weeks and say why the simple mean of the averages can be used. 3

OR

A final score combines Maths (score 42, weight 2), Science (score 58, weight 2). Find the weighted average score. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. The mean of 12 observations is 50. A new observation 63 is added. Find the new mean. 5

OR

20 mL of a 40% sugar solution is mixed with 60 mL of a 28% sugar solution. Find the percentage concentration of the mixture. 5

20. The mean of 10 observations is 35. A new observation 57 is added. Find the new mean. 5

OR

60 mL of a 28% sugar solution is mixed with 20 mL of a 24% sugar solution. Find the percentage concentration of the mixture. 5

END OF QUESTION PAPER — Total: 40 Marks